

CS2704

Group member: Jeremy Qiao(#3753194)

GitHub link: <https://github.com/Jeremy4/DataAnalysis/tree/main>

Explain the hypothesis:

- Current resident and healthcare personnel COVID-19 vaccination rates. Data are presented as one row per provider. On January 1, 2024, the CDC changed the data collection methodology for the percent of staff who are up to date with their COVID-19 vaccination. It may take facilities some time to adapt to the new methodology. As a result, the reported percent of staff who are up to date with their COVID-19 vaccination should be viewed with caution over the next few weeks. Consumers or interested parties should contact facilities directly for more information on their vaccination levels.
- State and National averages for facility resident and healthcare personnel COVID-19 vaccination rates. Data are presented as one row per state or territory plus a row for national averages.

Plan for testing your hypothesis:

Null Hypothesis (H_0): There is no correlation between resident and staff vaccination rates across facilities.

Alternative Hypothesis (H_1): There is a correlation between resident and staff vaccination rates across facilities.

2. Identify variables.

Percent of residence who are up to date on their vaccines

Percent of staff who are up to date on their vaccines

3. Choose statistical tests:

<https://data.cms.gov/provider-data/dataset/pvax-cv19>

4. Implement testing

Average Test

Heatmap Test

Z-Value and P-Value Test

P-Distribution

Linear regression line (Predictive module)

5. Interpret Results